

zh3r0 2020 Crypto - Mix

deuterium — 2020-06-18

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/zh3r0/crypto/Mix/2020-06-18-zh3r0-2020-Mix.html>

Abstract

CTF writeup: zh3r0 CTF 2020's Mix serves Base65536 with keyboard-shift clues and a Spoon esolang finish. Decoding comes with cutlery.

Mix

Description

At the BASEment no. 65536, A man is irritated with SHIFT key in his KEYBOARD as it's a sticky key, A kid is having chocolate icecream with a SPOON.

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Challenge Files

- [Mix.zip](#)
 - [Mix/chall'encrypted.txt](#)
 - [Mix/flag.txt](#)

[flag.txt](#) reads

If you opened this then you are a n00b

HAHA VERY FUNNY

[chall'encrypted.txt](#) contains some seemingly chinese.

since BASE65536 is a **BIG** hint to [base 65536](#), We do stuff.

```
import base65536
with open('chall_encrypted.txt','r') as chall_enc:
    data = chall_enc.read()

data1 = "".join( chr(ord(i)%256)+chr(ord(i)//256) for i in data )

d = base65536.decode(data)
print(d.decode())
```

Which prints out some neat stuff

flag 1:
Yjod od s lrunpstf djogy voljrtyrcy jrtr od upi g;sh xj4t-}U-i+dit4+

30303131303130302030313131303031302030303131303031312030313031313131312030303131303130
30203031313130313131203030313130303131203031313130303131203030313130303030203031313031
31303120303031313030313120303130313131313120303131313031313120303131303130303120303031
3130313131203031313031303030

[illegible]

Now we have 3 parts, lets work with hints as usual. So the first part must be KEYBOARD SHIFT cipher

At this point, I am wondering why is dcode.fr *SO GOOD*.



Search for a tool

★ SEARCH A TOOL ON DCODE BY KEYWORDS:

e.g. type sudoku

Results

!!	!!
qwerty →	This is a keyboard shift ciphertext here is you flag zh3r0{Y0u_sur3_
dworak ↵	Ikoī ed s 1cusntg ijegp zo0jltprry jltc ed uyi gqsd bj3ta}E-d+iin4?
dworak ↵	%koi ed s 1cusntg ijegp zo0jltprry jltc ed uyi gqsd bj3ta}E-d+iin4?
qwerty ↵	Hh9f ie x otjb0dgd fmitu c9]me5ufx6 me5t ie jo8 blw]j zu50+I[u]fkrv!
dworak →	Pqai ai n rcet.nhy iqafp wa0qchpcgp qchc ai e.u fznd kq3hs{Esu?iuh3?
qwerty →	Tupc 9f a ;fyh[xrr ch9hh fp'h4yhedu h4yf 9f y0o fpdn skrr'QJ08!cu55}
qwerty →	Uuic 9s d kfihoxyr ck9fh fi'k4rhtdt k4rf 9s i0u hpan shry' }j=8_co53}
qwerty ↵	Hk9s pe x oejm0agg smptt b9pmt5ftv6 mt5e pe j[8 b'wg cu3g=+Y[olskyv_
dworak ↵	Ijod od s 1rumpstf djogy xj4t-jU-i+d+it+4

KEYBOARD SHIFT CIPHER

Cryptography › Substitution Cipher › Keyboard Shift Cipher

KEYBOARD SHIFT DECODER

★ KEYBOARD SHIFTED CIPHERTEXT

Yjod od s 1rumpstf djogy v0ljrtqrcy jrtr od upi g;sh
xj4t-jU-i+d+it+4

★ PLAINTEXT EXPECTED LANGUAGE

★ KEYBOARD LAYOUT

★ SHIFT

★ USE ONLY ALPHANUMERIC CHARACTERS ☐

See also: [Keyboard Coordinates – Caesar Cipher](#)

KEYBOARD SHIFT ENCODER

★ KEYBOARD SHIFT PLAIN TEXT

dCode Keyboard

★ KEYBOARD LAYOUT ☒ AZERTY (FRENCH)
☐ QWERTY (US)
☐ DWORAK (US)

★ SHIFT

★ USE ONLY ALPHANUMERIC CHARACTERS ☐

See also: [Keyboard Coordinates – Caesar Cipher](#)

Tool to simulate a keyboard shift. Keyboard Shifting Cipher consists in typing a letter close to another on a computer keyboard. The shift can be on the right, the left, up or down.

ANSWERS TO QUESTIONS

 Lire en Français

Summary

- ★ Keyboard Shift Decoder
- ★ Keyboard Shift Encoder
- ★ How to encrypt using Keyboard Shifting cipher?
- ★ How to decrypt Keyboard Shifting cipher?
- ★ How to recognize Keyboard Shifting ciphertext?
- ★ What are the variants of the Keyboard Shifting cipher?

Similar tools

- ★ Caesar Cipher
- ★ Keyboard Coordinates
- ★ Letter Number (A1Z26) A=1, B=2, C=3
- ★ Morse Code
- ★ Mono-alphabetic Substitution
- ★ ROT Cipher
- ★ Binary Code
- ★ Polybius Cipher
- ★ Enigma Machine
- ★ T9 (Text Message)
- ★ All Tools ★

Support

- ★ Paypal
- ★ Patreon
- ★ More

Forum/Help

And we get the first part as This is a keyboard shift ciphertext here is you flag
zh3r0{Y0u_sur3_

Now moving to second part, we dont even need a hint,

```
f2 =
"3030313130313030203031313130303130203030313130303131203031303131313131203030313130313
03020303131313031313120303031313030313120303131313030313120303031313030303020303131303
13130312030303131303031312030313031313131312030313131303131312030313130313030312030303
13130313131203031313031303030"
print(bytes.fromhex(f2))

# b'00110100 01110010 00110011 01011111 00110100 01110111 00110011 01110011 00110000
01101101 00110011 01011111 01110111 01101001 00110111 01101000'

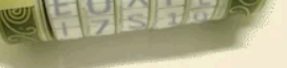
flag2 = "".join(chr(int(i,2)) for i in bytes.fromhex(f2).split())
print(flag2)
```

And just like that, we have 4r3_4w3s0m3_wi7h as second part of flag

Coming to third part, it seemingly is base64 encoding,

[illegible]

Which turns out to be some sort of binary, lets take third hint SPOON to our help and hey, never to forget dcode.fr



dCode

Search for a tool

★ SEARCH A TOOL ON dCODE BY KEYWORDS:

Results

📄
🔍
🖨️
⬇️
✕

Console

```
_411_7h3_ski115}
```

Memory: 1 => 20 ()
 2 => 30 ()
 3 => 53 (5)
 4 => 125 {}

Spoon - [dCode](#)

Tag(s) : Programming Language

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dCode and you

dCode is free and its tools are a valuable help in games, maths, geocaching, puzzles and problems to solve every day!

A suggestion ? a feedback ? a bug ? an idea ? *Write to dCode!*

Lire en Français

Summary

- ★ Spoon Interpreter
- ★ Spoon Encoder
- ★ How to encrypt using Spoon code?
- ★ How to decrypt Spoon code?
- ★ How to recognize Spoon coded text?

Similar tools

- ★ Brainfuck
- ★ Binary Code
- ★ ASCII Code
- ★ Javascript Unobfuscator
- ★ Ook!
- ★ Javascript Keycodes
- ★ Pikalang
- ★ Alphuck
- ★ ReverseFuck
- ★ Deadfish Language
- ★ All Tools ★

Support

- ★ Paypal
- ★ Patreon
- ★ More

Forum/Help

 DISCORD

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We get the third part of flag `_411_7h3_ski115}`

zh3r0-Y0u'sur3'4r3'4w3s0m3'wi7h'411'7h3'ski115"

Interactive controls are omitted from this PDF. See the canonical HTML version.